

DDR System v4.0 — YAML Representation Fidelity Report

Confirms comprehensive and accurate mapping of all normative content from `DDR System(Opus_v4).md` to `ddr_system_v4.0.yaml`.

Mapping Table

Source Reference	Source Section	YAML Key	Completeness
Design Philosophy (3 principles)	§1	<code>system_metadata.design_philosophy</code>	☐ All 3 principles with full descriptions
Changes from v3.1.1 (8 areas)	§1.1	<code>system_metadata.changes_from_prior</code>	☐ All 8 change areas with prior/current/rationale
Foundational Axioms (AX-1–AX-7)	§2	<code>axioms</code>	☐ All 7 axioms with statement and implication
Node schema properties (10 fields)	§3.1	<code>node_schema_fields</code>	☐ All 10 fields: <code>id</code> , <code>tier</code> , <code>title</code> , <code>content</code> , <code>parent_ids</code> , <code>status</code> , <code>version</code> , <code>created</code> , <code>modified</code> , <code>extension_annotations</code>
Edge type definitions (4 types)	§3.2	<code>edge_type_definitions.types</code>	☐ All 4 types with symbol and semantics
Edge type design decision	§3.2	<code>edge_type_definitions.design_decision</code>	☐ Verbatim rationale for 6→4 reduction
Universal node format	§3.3	<code>\$defs/DdrNode (schema) + nodes</code> structure	☐ Encoded in schema definition and each node entry
Core DAG topology (diagram)	§3.4	<code>nodes[].parent_ids</code> with <code>edge_types</code>	☐ All 9 topology edges encoded: <code>XPD→SIL(derives)</code> , <code>SIL→GPCL(derives)</code> , <code>GPCL→FCL(derives)</code> , <code>FCL→CL(derives)</code> , <code>FCL→SAL(derives)</code> , <code>CL→SAL(constrains)</code> , <code>SAL→ICL(derives)</code> , <code>ICL→CDL(implements)</code> , <code>CDL→ISL(implements)</code>
DAG invariants (6)	§3.5	<code>dag_invariants</code>	☐ All 6 invariants including C-2 <code>FCL→SAL</code> exception
Node ID format and immutability	§3.6	<code>node_id_format</code>	☐ General pattern, XPD pattern, examples, immutability rule
Citation rules (CIT-R1–CIT-R5)	§3.7	<code>citation_rules</code>	☐ All 5 citation rules

Source Reference	Source Section	YAML Key	Completeness
Consumption modes (2 modes)	§4	consumption_modes	☐ Both Express and Full modes with best-fit guidance
Express Mode groups (G1–G4)	§4	express_mode.groups	☐ All 4 groups with tier membership and labels
UNBUNDLE determinism rule	§4	express_mode.unbundle_determinism_rule	☐ Full rule text including rejection policy
XPD tier spec + 6 inclusion + 3 exclusion rules	§5 Tier 0	tier_definitions[tier_id=XPD]	☐ 9/9 rules
SIL tier spec + 6 inclusion + 4 exclusion rules	§5 Tier 1	tier_definitions[tier_id=SIL]	☐ 10/10 rules
GPCL tier spec + 10 inclusion + 3 exclusion rules	§5 Tier 2	tier_definitions[tier_id=GPCL]	☐ 13/13 rules + ORL absorption design decision
FCL tier spec + 6 inclusion + 3 exclusion rules	§5 Tier 3	tier_definitions[tier_id=FCL]	☐ 9/9 rules
CL tier spec + 10 inclusion + 3 exclusion rules	§5 Tier 4	tier_definitions[tier_id=CL]	☐ 13/13 rules + HIL/TDL unification design decision
SAL merge-node spec + 6 inclusion + 3 exclusion rules	§5 Tier 5	tier_definitions[tier_id=SAL]	☐ 9/9 rules; is_merge_node: true
ICL tier spec + 7 inclusion + 3 exclusion rules	§5 Tier 6	tier_definitions[tier_id=ICL]	☐ 10/10 rules
CDL tier spec + 7 inclusion + 3 exclusion rules	§5 Tier 7	tier_definitions[tier_id=CDL]	☐ 10/10 rules
ISL tier spec + 6 inclusion + 2 exclusion rules	§5 Tier 8	tier_definitions[tier_id=ISL]	☐ 8/8 rules; is_terminal_leaf: true

Source Reference	Source Section	YAML Key	Completeness
Constraint precedence (9-tier table)	§6	<code>constraint_precedence.tiers</code>	☐ All 9 priorities with rationale
Override principle (XPD veto)	§6	<code>constraint_precedence.override_principle</code>	☐ Absolute veto right language preserved
Intra-tier conflict rule	§6	<code>constraint_precedence.intra_tier_conflict_rule</code>	☐ Full rule text
Physical constraint escalation	§6	<code>constraint_precedence.physical_constraint_escalation</code>	☐ Full escalation policy
7 core operations	§7.1	<code>operations.core_operations</code>	☐ All 7: INSERT, DELETE, MODIFY, SUPERSEDE, VERIFY, VALIDATE, UNBUNDLE
Removed operations design decision	§7.1	<code>operations.design_decision_removed_ops</code>	☐ RELOCATE/ABSTRACT/CONCRETIZE/DETECT_* removal rationale
Dirty flag triggers (5)	§7.2	<code>operations.dirty_flag_triggers</code>	☐ All 5 triggers with scope
Dirty flag notes (4)	§7.2	<code>operations.dirty_flag_notes</code>	☐ Node insertion, supersede auto-update, supersede-MODIFY interaction, deprecation lifecycle
Resolution workflow	§7.3	<code>operations.resolution_workflow</code>	☐ Full workflow sequence
Reconciliation manifest fields	§7.3	<code>operations.reconciliation_manifest_tracks</code>	☐ All 5 tracked items
Extension architecture (permitted/prohibited)	§8.1	<code>extension_system.permitted_actions / .prohibited_actions</code>	☐ 4 permitted, 4 prohibited
Extension Candidate Pool (ARE)	§8.2	<code>extension_system.candidate_pool</code>	☐ All 6 candidate pool properties
Extension integration rules (EXT-R1–EXT-R7)	§8.3	<code>extension_system.integration_rules</code>	☐ All 7 rules
EXT-R2 normative note	§8.3	<code>extension_system.normative_notes</code>	☐ "All Core tiers" invalidity note
E1 HRE (4 rules)	§9	<code>extension_catalog[id=E1]</code>	☐ contract, reads[4], annotates[2], 4 rules
E2 DGA (3 rules)	§9	<code>extension_catalog[id=E2]</code>	☐ contract, reads[4], annotates[2], 3 rules
E3 LVE (4 rules)	§9	<code>extension_catalog[id=E3]</code>	☐ contract, reads[9], annotates[9], 4 rules
E4 ORE (4 rules)	§9	<code>extension_catalog[id=E4]</code>	☐ contract, reads[5], annotates[2], 4 rules

Source Reference	Source Section	YAML Key	Completeness
E5 ARE (4 rules + annotation restriction note)	§9	extension_catalog[id=E5]	☐ contract, reads[4], annotates[4], 4 rules, notes
E6 SCE (5 rules)	§9	extension_catalog[id=E6]	☐ contract, reads[4], annotates[3], 5 rules
E7 DDE (4 rules + FCL annotation rationale)	§9	extension_catalog[id=E7]	☐ contract, reads[5], annotates[3], 4 rules, notes (resolves Audit M-1)
E8 DCP (4 rules)	§9	extension_catalog[id=E8]	☐ contract, reads[3], annotates[2], 4 rules
E9 EHD (5 rules incl. synthetic XPD rule)	§9	extension_catalog[id=E9]	☐ contract, reads[5], annotates[3], 5 rules including EHD-R5 synthetic XPD-equivalent
Compliance checklist §11 (8+9+4 items)	§11	compliance_checklist	☐ 8 structural, 9 atomic rule, 4 extension items
Glossary (12 terms)	Glossary	glossary	☐ All 12 terms: Atomic Rule, Candidate Pool, DAG, Dirty Flag, Edge Type, Express Mode, Extension, Leaf Node, Merge Node, Orphan, Root Node, Tier Contamination
Version history (5 entries)	Appendix A	version_history	☐ v1.0 through v4.0
Tier migration map (11 entries)	Appendix B	tier_migration.tier_map	☐ All 11 tier mappings
Rule migration cross-reference (14 entries)	Appendix B	tier_migration.rule_map	☐ All 14 rule mappings; Audit C-3 (ORL-R7 TBD) resolved with annotation
Migration policy statement	Appendix B	tier_migration.policy	☐ Verbatim policy text
Canonical 9-node DAG topology	§3.4 diagram	nodes (9 entries)	☐ All 9 tier nodes with canonical topology edges and substantive content

Rule Count Verification

Tier	Inclusion Rules	Exclusion Rules	Total in Spec	Total in YAML
XPD	6 (R1–R6)	3 (E1–E3)	9	☐ 9
SIL	6 (R1–R6)	4 (E1–E4)	10	☐ 10

	Tier	Inclusion Rules	Exclusion Rules	Total in Spec	Total in YAML
	GPCL	10 (R1–R10)	3 (E1–E3)	13	☐ 13
	FCL	6 (R1–R6)	3 (E1–E3)	9	☐ 9
	CL	10 (R1–R10)	3 (E1–E3)	13	☐ 13
	SAL	6 (R1–R6)	3 (E1–E3)	9	☐ 9
	ICL	7 (R1–R7)	3 (E1–E3)	10	☐ 10
	CDL	7 (R1–R7)	3 (E1–E3)	10	☐ 10
	ISL	6 (R1–R6)	2 (E1–E2)	8	☐ 8
	Total	64	27	91	☐ 91/91

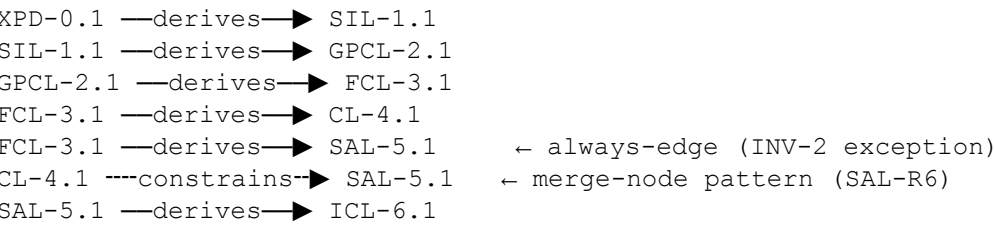
Audit Finding Resolutions

The following issues identified in the prior Logic Audit (Prompt #2) were actively resolved during YAML authoring:

Audit ID	Finding	Resolution in YAML
C-2	FCL→SAL tier-skip exception undocumented	<code>dag_invariants[id=INV-2]</code> explicitly encodes the exception with rationale
C-3	ORL-R7 mapping marked TBD in Finalized doc	<code>tier_migration.rule_map</code> assigns ORL-R7→GPCL-R10 with annotation flagging pending board confirmation
H-1	No formal status transition model	<code>\$defs/DdrNode.status</code> description enumerates all 10 valid transitions with triggering operations
M-1	DDE FCL annotation lacked rationale	<code>extension_catalog[id=E7].notes</code> adds explicit rationale per recommendation

Structural Verification: Canonical Nodes DAG

The 9 canonical nodes encode the exact topology from §3.4:



ICL-6.1 —implements→ CDL-7.1
CDL-7.1 —implements→ ISL-8.1

Verified: no cycles; no tier-skipping (FCL→SAL exception encoded); all non-root nodes have ≥ 1 parent_id; SAL-5.1 carries both required parents; ISL-8.1 is the sole leaf node; XPD-0.1 is the sole root.